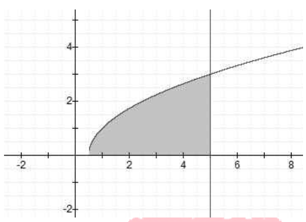
1. 4번	2. 2번	3. 2번	4. 3번	5. 2번
6. 2번	7. 4번	8. 3번	9. 2번	10. 1번
11. 4번	12. 4번	13. 1번	14. 1번	15. 1번
16. 3번	17. 2번	18. 3번	19. 3번	20. 4번
21. 3번	22. 4번	23. 3번	24. 2번	25. 4번
26. 3번	27. 3번	28. 4번	29. 3번	30. 3번
31. 4번	32. 2번	33. 3번	34. 5번	35. 2번
36. 5번	37. 1번	38. 4번	39. 2번	40. 4번
41. 1번	42. 2번	43. 2번	44. 2번	45. 1번
46. 2번	47. 1번	48. 4번	49. 1번	50. 4번
51. 4번	52. 3번	53. $\sqrt{2}$		

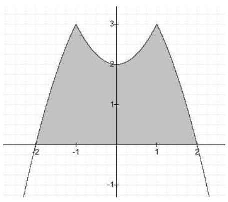
1.



곡선 $y=\sqrt{2x-1}$ 은 $\left(\frac{1}{2},0\right)$을 지나므로 적분 구간은 $\left[\frac{1}{2},5\right]$ 이다.

즉, $\int_{\frac{1}{2}}^{5}\sqrt{2x-1}\,dx=\frac{1}{2}\times\frac{2}{3}\left[(2x-1)^{\frac{3}{2}}\right]_{\frac{1}{2}}^{5}=9$

2.



$y=\begin{cases}x^2+2 & (-1<x<1)\\ -x^2+4 & (x<-1\ ,\ x>1)\end{cases}$ 이고 $(-2,0),(2,0)$을 지나므로

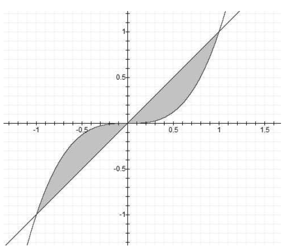
$y=3-|x^2-1|$와 $y=0$으로 둘러싸인 도형의 넓이는

$$\int_{-2}^{2}|y-0|dx=\int_{-2}^{2}3-|x^2-1|dx\ \ (\because \text{구간}[-2,2]\text{에서 } y\text{는 항상 양수})$$

$$=2\left(\int_0^1 x^2+2\,dx+\int_1^2 -x^2+4dx\right)\ \ (\because \text{우함수})$$

$$=2\left(\left[\frac{1}{3}x^3+2x\right]_0^1+\left[-\frac{1}{3}x^3+4x\right]_1^2\right)=8$$

3.



$f(x)$와 $g(x)$는 $(-1,1),(0,0),(1,1)$에서 만나고

구간 $[-1,0]$에서 $f(x)>g(x)$, 구간 $[0,1]$에서 $g(x)>f(x)$이므로

영역의 넓이는

$$\int_{-1}^{1}|f(x)-g(x)|dx$$

$$=\int_{-1}^{0}f(x)-g(x)dx+\int_0^1 g(x)-f(x)dx$$

$$=\int_{-1}^{0}x^3-xdx+\int_0^1 x-x^3dx$$

$$=\int_0^1 t-t^3dt+\int_0^1 x-x^3dx\ \ (\because x=-t\text{치환})$$

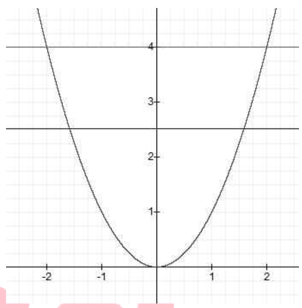
$$=2\int_0^1 x-x^3dx=2\left[\frac{1}{2}x^2-\frac{1}{4}x^4\right]_0^1=\frac{1}{2}$$

[참고]

둘러싸인 영역의 면적은 원점을 기준으로 대칭이므로

한쪽 영역의 넓이 $\times 2$를 해도 된다.

4.



곡선 $y=x^2$은 $(-2,4),(2,4)$를 지나므로 구간은 $[-2,2]$이고,

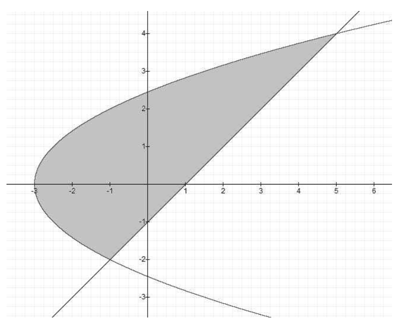
우함수이므로 구간 $[0,2]$에서만 확인해도 된다.

이때, 곡선이 $(\sqrt{b},b)$를 지나므로

$$\int_0^2 4-x^2dx=2\int_0^{\sqrt{b}}b-x^2dx$$

$\Leftrightarrow\ \frac{16}{3}=\frac{4}{3}b\sqrt{b}$ 이므로 $b=4^{\frac{2}{3}}$ 이다.

5.



두 함수가 만나는 점을 찾기 위해 직선 $y=x-1$을

포물선 $y^2=2x+6$에 대입하면

$(x-1)^2=2x+6$

$\Leftrightarrow\ x^2-4x-5=0$

$\Leftrightarrow\ (x-5)(x+1)=0$에서 $x=-1$또는5이다.

따라서 두 함수가 만나는 점은 $(-1,-2),(5,4)$이다.

이제 편의를 위해 $x=y+1,\ x=\frac{y^2-6}{2}$ 으로 바꾸면

둘러싸인 영역의 넓이는

$$\int_{-2}^{4}(y+1)-\left(\frac{1}{2}y^2-3\right)dy=\int_{-2}^{4}-\frac{1}{2}y^2+y+4dy$$

$$=\left[-\frac{1}{6}y^3+\frac{1}{2}y^2+4y\right]_{-2}^{4}=18$$

6. 곡선 $x=\cos^3 t, y=\sin^2 t$는

$t=\pi$일 때 $(-1,0)$, $t=0$일 때 $(1,0)$을 지나므로

곡선과 x축으로 둘러싸인 부분의 넓이는

$$\int_{-1}^{1}|y|\,dx=\int_{\pi}^{0}|\sin^2 t|\,3\cos^2 t\times(-\sin t)dt$$

$$=3\int_0^{\pi}\cos^2 t\sin^3 t dt=3\int_0^{\pi}\sin^3 t-\sin^5 t dt$$

$$=3\left(2\times\frac{2}{3}\times1-2\times\frac{4}{5}\times\frac{2}{3}\times1\right)=\frac{4}{5}\quad(\because Wallis\ \text{정리})$$

7. 싸이클로이드의 넓이$=3\pi a^2$이므로 27π이다.

[다른 풀이]

곡선 $x=3(\theta-\sin\theta), y=3(1-\cos\theta)$는 $\theta=0$일 때 $(0,0)$

$\theta=2\pi$일 때 $(6\pi,0)$을 지나므로

곡선과 x축으로 둘러싸인 부분의 넓이는

$$\int_0^{6\pi}|y|\,dx=\int_0^{2\pi}|3(1-\cos\theta)|\times3(1-\cos\theta)d\theta$$

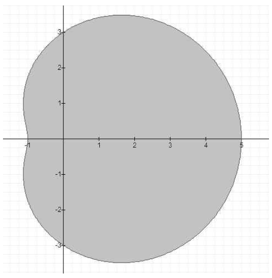
$$=9\int_0^{2\pi}(1-\cos\theta)^2 d\theta\quad(\because 1-\cos\theta\text{는}0\le\theta\le2\pi\text{에서 항상 양수})$$

$$=9\int_0^{2\pi}1-2\cos\theta+\cos^2\theta d\theta$$

$$=9\times2\pi+0+9\times4\times\frac{1}{2}\times\frac{\pi}{2}\quad(\because Wallis\ \text{정리})$$

$$=27\pi$$

8.



$$\frac{1}{2}\int_0^{2\pi}r^2 d\theta=\frac{1}{2}\int_0^{2\pi}(2\cos\theta+3)^2 d\theta$$

$$=2\times\frac{1}{2}\int_0^{\pi}(2\cos\theta+3)^2 d\theta\quad(\because x\text{축 대칭})$$

$$=\int_0^{\pi}4\cos^2\theta+12\cos\theta+9d\theta$$

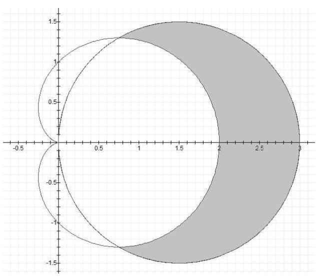
$$=4\times2\times\frac{1}{2}\times\frac{\pi}{2}+0+9\pi=11\pi$$

[다른 풀이]

$a=3$, $b=2$이므로 영역의 넓이는 공식에 의하여

$$\frac{\pi}{2}(2a^2+b^2)=\frac{\pi}{2}(18+4)=11\pi$$

9.



$3\cos\theta=1+\cos\theta \Leftrightarrow \theta=\frac{\pi}{3},\frac{5}{3}\pi$ 이므로 넓이는

$$\frac{1}{2}\int_0^{\frac{\pi}{3}}r_1^2-r_2^2 d\theta\ +\ \frac{1}{2}\int_{\frac{5}{3}\pi}^{2\pi}r_1^2-r_2^2 d\theta$$

$$=2\times\frac{1}{2}\int_0^{\frac{\pi}{3}}r_1^2-r_2^2 d\theta\quad(\because \text{대칭성})$$

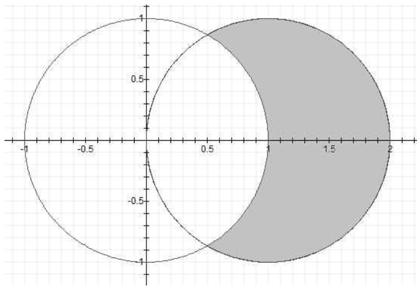
$$=2\times\frac{1}{2}\int_0^{\frac{\pi}{3}}9\cos^2\theta-(1+\cos\theta)^2 d\theta=\int_0^{\frac{\pi}{3}}-1-2\cos\theta+8\cos^2\theta\,d\theta$$

$$=\int_0^{\frac{\pi}{3}}3-2\cos\theta+4\cos2\theta\,d\theta=[3\theta-2\sin\theta+2\sin2\theta]_0^{\frac{\pi}{3}}=\pi$$

[다른 풀이]

$a=1$이므로 $r=3\cos\theta$의 내부와 $r=1+\cos\theta$의 외부로 둘러싸인 영역의 넓이는 공식 πa^2 에 의하여 π이다.

10.

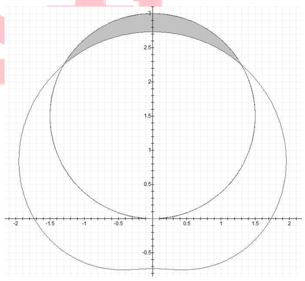


$$S=2\times\frac{1}{2}\int_0^{\frac{\pi}{3}}4\cos^2\theta-1\,d\theta$$

$$=\int_0^{\frac{\pi}{3}}1+2\cos2\theta\,d\theta$$

$$=[\theta+\sin2\theta]_0^{\frac{\pi}{3}}=\frac{\pi}{3}+\frac{\sqrt{3}}{2}$$

11.

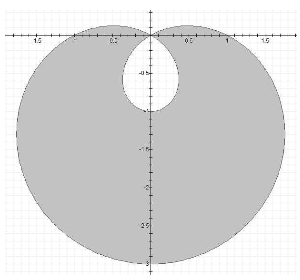


$$S=2\times\frac{1}{2}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}9\sin^2\theta-\left(\sqrt{3}+\sin\theta\right)^2 d\theta$$

$$=\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}8\sin^2\theta-3-2\sqrt{3}\sin\theta\,d\theta=\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}1-4\cos2\theta-2\sqrt{3}\sin\theta\,d\theta$$

$$=[\theta-2\sin2\theta+2\sqrt{3}\cos\theta]_{\frac{\pi}{3}}^{\frac{\pi}{2}}=\frac{\pi}{6}$$

12.



극방정식 $r=1-2\sin\theta$과 $r=1-2\cos\theta$의 내부곡선과 외부곡선으로 둘러싸인 부분의 면적은 같다.

이때, 내부 곡선의 넓이는 $2\times\frac{1}{2}\int_0^{\frac{\pi}{3}}(1-2\cos\theta)^2 d\theta$

외부 곡선의 넓이는 $2\times\frac{1}{2}\int_{\frac{\pi}{3}}^{\pi}(1-2\cos\theta)^2 d\theta$이므로

둘러싸인 부분의 면적은

$$2\times\left[\frac{1}{2}\int_{\frac{\pi}{3}}^{\pi}(1-2\cos\theta)^2\,d\theta-\frac{1}{2}\int_0^{\frac{\pi}{3}}(1-2\cos\theta)^2\,d\theta\right]$$

$$=\int_{\frac{\pi}{3}}^{\pi}(1-2\cos\theta)^2\,d\theta-\int_0^{\frac{\pi}{3}}(1-2\cos\theta)^2\,d\theta$$

$$=\int_{\frac{\pi}{3}}^{\pi}(3-4\cos\theta+2\cos2\theta)\,d\theta-\int_0^{\frac{\pi}{3}}(3-4\cos\theta+2\cos2\theta)\,d\theta$$

$$=[3\theta-4\sin\theta+\sin2\theta]_{\frac{\pi}{3}}^{\pi}-[3\theta-4\sin\theta+\sin2\theta]_0^{\frac{\pi}{3}}$$

$$=\pi+3\sqrt{3}$$

[다른 풀이]

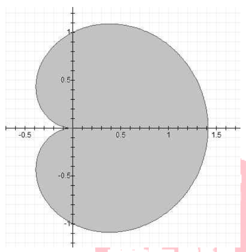
$r=1-2\sin\theta$과 $r=1+2\cos\theta$의 내부곡선과 외부곡선으로 둘러싸인 부분의 면적은 같다.

따라서 공식에 의하면 $r=a(1+2\cos\theta)$의 내부곡선과 외부곡선으로 둘러싸인 부분의 면적은 $(\pi+3\sqrt{3})a^2$이므로 $S=\pi+3\sqrt{3}$이다.

13. $1+2\sin\theta=0 \iff \sin\theta=-\frac{1}{2}$이므로 $\theta=\frac{7\pi}{6},\ \frac{11\pi}{6}$이다.

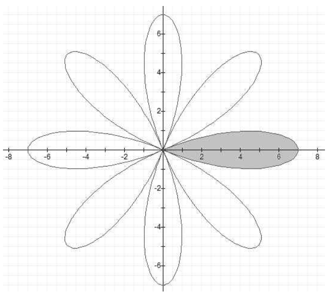
즉, $\frac{1}{2}\int_{\frac{7}{6}\pi}^{\frac{11}{6}\pi}(1+2\sin\theta)^2\,d\theta$

14.



$$\frac{1}{2}\int_0^{2\pi}1+\cos\theta\,d\theta=\frac{1}{2}[\theta+\sin\theta]_0^{2\pi}=\pi$$

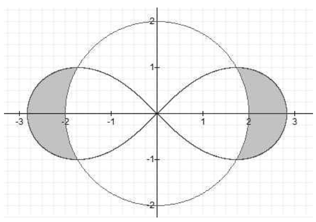
15.



$$2\times\frac{1}{2}\int_0^{\frac{\pi}{8}}49\cos^2 4\theta\,d\theta=49\int_0^{\frac{\pi}{8}}\frac{1+\cos8\theta}{2}\,d\theta$$

$$=\frac{49}{2}\left[\theta+\frac{1}{8}\sin8\theta\right]_0^{\frac{\pi}{8}}=\frac{49}{16}\pi$$

16.



$$4\times\frac{1}{2}\int_0^{\frac{\pi}{6}}8\cos2\theta-4\,d\theta=2[4\sin2\theta-4\theta]_0^{\frac{\pi}{6}}=4\sqrt{3}-\frac{4}{3}\pi$$

17. $\int_1^2\sqrt{1+\left(\frac{1}{2}x^3-\frac{1}{2x^3}\right)^2}\,dx=\int_1^2\sqrt{\left(\frac{1}{2}x^3+\frac{1}{2x^3}\right)^2}\,dx$

$$=\int_1^2\frac{1}{2}x^3+\frac{1}{2x^3}\,dx=\left[\frac{1}{8}x^4-\frac{1}{4x^2}\right]_1^2=\frac{33}{16}$$

18. $\int_1^9\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy=\int_1^9\sqrt{1+\left(\frac{\sqrt{y}}{2}-\frac{1}{2\sqrt{y}}\right)^2}\,dy$

$$=\int_1^9\sqrt{\left(\frac{\sqrt{y}}{2}+\frac{1}{2\sqrt{y}}\right)^2}\,dy=\int_1^9\frac{\sqrt{y}}{2}+\frac{1}{2\sqrt{y}}\,dy$$

$$=\int_1^9\frac{\sqrt{y}}{2}+\frac{1}{2\sqrt{y}}\,dy=\left[\frac{1}{3}y^{\frac{3}{2}}+\sqrt{y}\right]_1^9=\frac{32}{3}$$

19. $y'=\sqrt{\cos2x}$이므로 곡선의 길이는

$$\int_0^{\frac{\pi}{4}}\sqrt{1+\cos2x}\,dx=\int_0^{\frac{\pi}{4}}\sqrt{2\cos^2x}\,dx$$

$$=\sqrt{2}\int_0^{\frac{\pi}{4}}\cos x\,dx=\sqrt{2}[\sin x]_0^{\frac{\pi}{4}}=1$$

20. $\int_0^1\sqrt{(\sin t+t\cos t)^2+(\cos t-t\sin t)^2}\,dt$

$$=\int_0^1\sqrt{1+t^2}\,dt=\int_0^{\frac{\pi}{4}}\sec^3\theta\,d\theta$$

$$=\frac{1}{2}[\sec\theta\tan\theta+\ln(\sec\theta+\tan\theta)]_0^{\frac{\pi}{4}}$$

$$=\frac{1}{2}[\sqrt{2}+\ln(\sqrt{2}+1)]$$

21. $\int_0^{\frac{\pi}{6}}\sqrt{(-3\cos^2t\sin t)^2+(3\sin^2t\cos t)^2+(4\sin t\cos t)^2}\,dt$

$$=\int_0^{\frac{\pi}{6}}\sqrt{25\sin^2t\cos^2t}\,dt=\int_0^{\frac{\pi}{6}}5\sin t\cos t\,dt=\frac{5}{2}[\sin^2t]_0^{\frac{\pi}{6}}=\frac{5}{8}$$

22. 싸이클로이드의 한 호의 길이는 $8a$이므로 곡선의 길이는 16이다.

[다른 풀이]

곡선의 길이는

$$\int_0^{2\pi}\sqrt{(2(1-\cos\theta))^2+(2\sin\theta)^2}\,d\theta$$

$$=\int_0^{2\pi}\sqrt{8-8\cos\theta}\,d\theta$$

$$=\int_0^{2\pi}4\sqrt{\frac{1-\cos\theta}{2}}\,d\theta$$

$$=4\int_0^{2\pi}\left|\sin\frac{\theta}{2}\right|\,d\theta$$

$$=4\int_0^{2\pi}\sin\frac{\theta}{2}\,d\theta$$

$$=\left[8\cos\frac{\theta}{2}\right]_0^{2\pi}=16$$

23. 성망형의 전체의 길이는 $6a$이므로 곡선의 길이는 6이다.

[다른 풀이]

$x=\cos^3t,\ y=\sin^3t$이므로 곡선의 길이는

$$\int_0^{2\pi}\sqrt{(-3\cos^2t\sin t)^2+(3\sin^2t\cos t)^2}\,dt$$

$$=\int_0^{2\pi}\sqrt{9\cos^2t\sin^2t(\cos^2t+\sin^2t)}\,dt$$

$$=\int_0^{2\pi}|3\cos t\sin t|\,dt$$

$$=\frac{3}{2}\int_0^{2\pi}|\sin2t|\,dt$$

$$=4\times\frac{3}{2}\int_0^{\frac{\pi}{2}}\sin2t\,dt=6\left[-\frac{1}{2}\cos2t\right]_0^{\frac{\pi}{2}}=6$$

24. $r=1-\sin\theta\ \left(0\le\theta\le\frac{\pi}{2}\right)$의 길이는

$r=1-\cos\theta\ \left(0\le\theta\le\frac{\pi}{2}\right)$의 길이와 같다.

$$\int_0^{\frac{\pi}{2}}\sqrt{(1-\cos\theta)^2+\sin^2\theta}\,d\theta=\int_0^{\frac{\pi}{2}}\sqrt{2-2\cos\theta}\,d\theta$$

$$=\int_0^{\frac{\pi}{2}}2\sin\frac{\theta}{2}d\theta=\left[-4\cos\frac{\theta}{2}\right]_0^{\frac{\pi}{2}}=4-2\sqrt{2}$$

25. $r=1-\cos\theta$와 $r=1+\cos\theta$의 곡선의 길이는 같고,
심장형 $r=a(1+\cos\theta)$의 전체 길이는 $8a$이므로 곡선의 길이는 8이다.

[다른 풀이]
곡선의 길이는

$$\int_0^{2\pi}\sqrt{(1-\cos\theta)^2+\sin^2\theta}\,d\theta$$

$$=\int_0^{2\pi}\sqrt{2-2\cos\theta}\,d\theta$$

$$=2\int_0^{2\pi}\sqrt{\frac{1-\cos\theta}{2}}\,d\theta$$

$$=2\int_0^{2\pi}\left|\sin\frac{\theta}{2}\right|d\theta$$

$$=2\int_0^{2\pi}\sin\frac{\theta}{2}d\theta$$

$$=4\left[-\cos\frac{\theta}{2}\right]_0^{2\pi}=8$$

26. $\int_0^1\sqrt{\theta^4+4\theta^2}\,d\theta=\int_0^1\theta\sqrt{\theta^2+4}\,d\theta=\left[\frac{1}{3}(\theta^2+4)^{\frac{3}{2}}\right]_0^1=\frac{1}{3}(5\sqrt{5}-8)$

27. $\int_0^{\pi}\sqrt{\sin^6\left(\frac{\theta}{3}\right)+\sin^4\left(\frac{\theta}{3}\right)\cos^2\left(\frac{\theta}{3}\right)}\,d\theta$

$$=\int_0^{\pi}\sqrt{\sin^4\left(\frac{\theta}{3}\right)}\,d\theta=\int_0^{\pi}\sin^2\left(\frac{\theta}{3}\right)d\theta=\frac{1}{2}\int_0^{\pi}1-\cos\left(\frac{2\theta}{3}\right)d\theta$$

$$=\frac{1}{2}\left[\theta-\frac{3}{2}\sin\left(\frac{2\theta}{3}\right)\right]_0^{\pi}=\frac{1}{2}\left(\pi-\frac{3}{4}\sqrt{3}\right)$$

28. Pappus 정리에 의하여 회전체의 부피는 $2\pi\times1\times\pi\times1^2$이므로 $2\pi^2$이다.

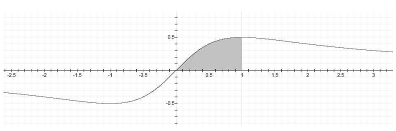
29. Pappus 정리에 의하여 회전체의 부피는 $2\pi\times3\times\pi\times2^2$이므로 $24\pi^2$이다.

30. Pappus 정리에 의하여 회전체의 겉넓이는 $2\pi\times4\times\pi\times2^2$이므로 $32\pi^2$이다.

31. Pappus 정리에 의하여 회전체의 부피는 $2\pi\times\frac{3}{\sqrt{2}}\times\frac{1}{2}\times2\times3$이므로
$9\sqrt{2}\pi$이다.

32. Pappus 정리에 의하여 회전체의 부피는 $2\pi\times\frac{3}{\sqrt{5}}\times\frac{1}{2}$이므로 $\frac{3}{\sqrt{5}}\pi$이다.

33.

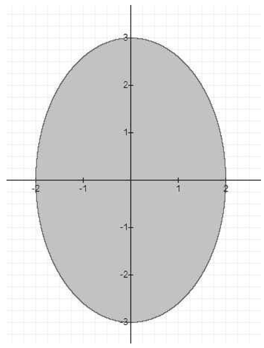


$$V_x=\pi\int_0^1 y^2dx=\pi\int_0^1\frac{x^2}{(x^2+1)^2}dx=\pi\int_0^{\frac{\pi}{4}}\frac{\tan^2\theta}{\sec^4\theta}\sec^2\theta d\theta$$

$$=\pi\int_0^{\frac{\pi}{4}}\sin^2\theta d\theta=\frac{\pi}{2}\int_0^{\frac{\pi}{4}}1-\cos2\theta d\theta=\frac{\pi}{2}\left[\theta-\frac{1}{2}\sin2\theta\right]_0^{\frac{\pi}{4}}$$

$$=\frac{\pi}{8}(\pi-2)$$

34.

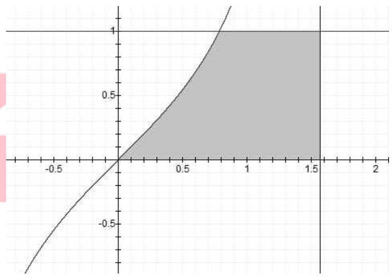


$$V_x=2\times\pi\int_0^2 y^2dx=2\pi\times9\int_0^2\left(1-\frac{x^2}{4}\right)dx$$

$$=18\pi\left[x-\frac{x^3}{12}\right]_0^2=24\pi$$

[다른 풀이]
타원 $\frac{x^2}{4}+\frac{y^2}{9}=1$로 둘러싸인 영역을 x축의 둘레로 회전시키면
회전체는 타원체가 된다.
이때 타원의 반지름은 x축 방향으로 2, y축 방향으로 3이므로,
회전하여 얻어지는 타원체의 반지름은 각각 $3,3,2$이다.
따라서 타원체의 부피는
$V=\frac{4}{3}\pi abc=\frac{4}{3}\pi\times3\times3\times2=24\pi$이다.

35.

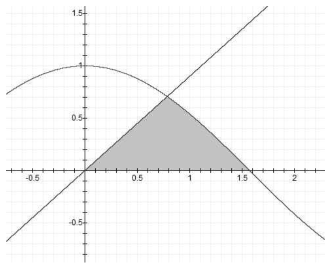


$$V_x=\pi\int_0^{\frac{\pi}{4}}\tan^2xdx+\pi\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}1dx$$

$$=\pi\int_0^{\frac{\pi}{4}}\sec^2x-1dx+\pi\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}1dx=\pi[\tan x-x]_0^{\frac{\pi}{4}}+\frac{\pi^2}{4}$$

$$=\pi\left(1-\frac{\pi}{4}\right)+\frac{\pi^2}{4}=\pi$$

36.

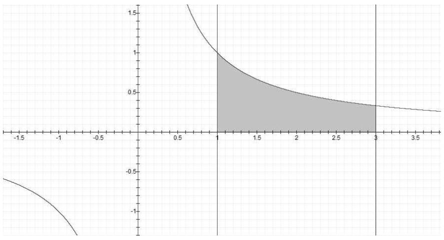


$$V_x=\pi\int_0^{\frac{\pi}{4}}\frac{8}{\pi^2}x^2dx+\pi\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\cos^2xdx$$

$$=\frac{8}{\pi}\int_0^{\frac{\pi}{4}}x^2dx+\pi\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\frac{1+\cos2x}{2}dx$$

$$=\frac{8}{\pi}\left[\frac{1}{3}x^3\right]_0^{\frac{\pi}{4}}+\frac{\pi}{2}\left[x+\frac{1}{2}\sin2x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}=\frac{\pi^2}{6}-\frac{\pi}{4}$$

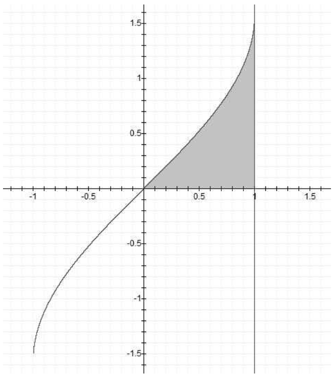
37.



$$V_x = \pi\int_1^3 \left(\frac{1}{x}+1\right)^2 - 1dx = \pi\int_1^3 \frac{1}{x^2}+\frac{2}{x}dx$$

$$= \pi\left[-\frac{1}{x}+2\ln x\right]_1^3 = 2\pi\left(\frac{1}{3}+\ln 3\right)$$

38.

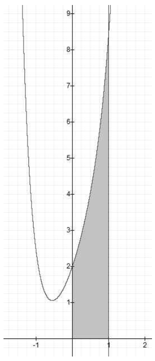


$$V = 2\pi\int_0^1 xydx = 2\pi\int_0^1 x\sin^{-1}xdx$$

$$= 2\pi\left[\frac{1}{2}x^2\sin^{-1}x\right]_0^1 - \pi\int_0^1 \frac{x^2}{\sqrt{1-x^2}}dx$$

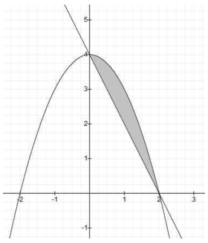
$$= \frac{\pi^2}{2} - \pi\int_0^{\frac{\pi}{2}}\sin^2\theta\, d\theta = \frac{\pi^2}{4}$$

39.



$$V = 2\pi\int_0^1 xydx = 2\pi\int_0^1 x\left(3x+2e^{x^2}\right)dx = 2\pi\left[x^3+e^{x^2}\right]_0^1 = 2\pi e$$

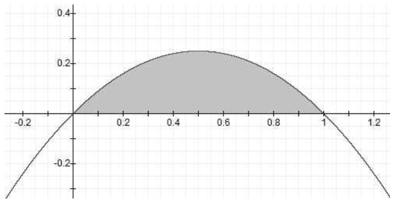
40.



$$V = 2\pi\int_0^2 x\left[(-x^2+4)-(-2x+4)\right]dx$$

$$= 2\pi\int_0^2 -x^3+2x^2dx = 2\pi\left[-\frac{1}{4}x^4+\frac{2}{3}x^3\right]_0^2 = \frac{8}{3}\pi$$

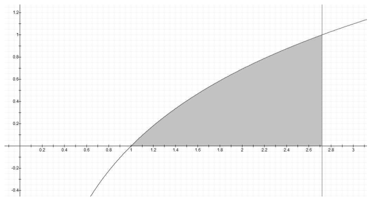
41.



$$V = 2\pi\int_0^1 (3-x)(x-x^2)dx = 2\pi\int_0^1 x^3-4x^2+3xdx$$

$$= 2\pi\left[\frac{1}{4}x^4-\frac{4}{3}x^3+\frac{3}{2}x^2\right]_0^1 = \frac{5}{6}\pi$$

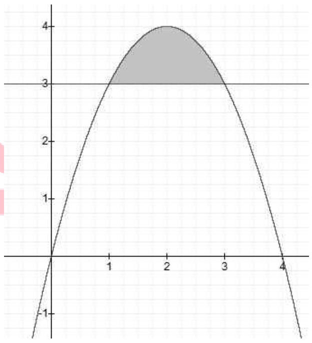
42.



$$V = 2\pi\int_1^e (x+1)\ln xdx = 2\pi\left[\left(\frac{1}{2}x^2+x\right)\ln x - \int\left(\frac{1}{2}x+1\right)dx\right]_1^e$$

$$= 2\pi\left[\left(\frac{1}{2}x^2+x\right)\ln x - \left(\frac{1}{4}x^2+x\right)\right]_1^e = \frac{\pi}{2}(e^2+5)$$

43.

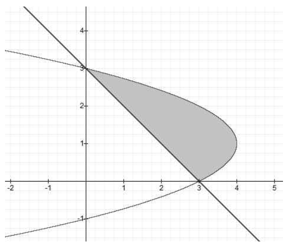


$$V = 2\pi\int_1^3 (x-1)\left[x(4-x)-3\right]dx$$

$$= 2\pi\int_1^3 -x^3+5x^2-7x+3dx$$

$$= 2\pi\left[-\frac{1}{4}x^4+\frac{5}{3}x^3-\frac{7}{2}x^2+3x\right]_1^3 = \frac{8}{3}\pi$$

44.



$$V = 2\pi\int_0^3 x\left[(-x^2+2x+3)-(3-x)\right]dx$$

$$= 2\pi\int_0^3 -x^3+3x^2dx = 2\pi\left[-\frac{1}{4}x^4+x^3\right]_0^3 = \frac{27}{2}\pi$$

45. $V = \int_0^1 \text{단면의 넓이}\, dx = \int_0^1 \frac{\sqrt{3}}{4}y^2dx = \frac{\sqrt{3}}{4}\int_0^1 x^6dx = \frac{\sqrt{3}}{28}$

46. $V = \int_0^2 \text{단면의 넓이}\, dy = \int_0^2 4x^2dy = 4\int_0^2 2-ydy = 4\left[2y-\frac{1}{2}y^2\right]_0^2 = 8$

47. $S_x = 2\pi\int_0^1 y\sqrt{1+(y')^2}\,dx = 2\pi\int_0^1 x^3\sqrt{1+9x^4}\,dx$

$= 2\pi\left[\frac{1}{36}\times\frac{2}{3}(1+9x^4)^{\frac{3}{2}}\right]_0^1 = \frac{\pi}{27}(10\sqrt{10}-1)$

48. $S_x = 2\pi\int_0^1 y\sqrt{1+(y')^2}\,dx = 2\pi\int_0^1 \sqrt{x}\sqrt{1+\frac{1}{4x}}\,dx$

$= 2\pi\int_0^1 \sqrt{x+\frac{1}{4}}\,dx = 2\pi\left[\frac{2}{3}\left(x+\frac{1}{4}\right)^{\frac{3}{2}}\right]_0^1 = \frac{\pi}{6}(5\sqrt{5}-1)$

49. $L = \int_1^{16}\sqrt{1+\sqrt{x}-1}\,dx = \int_1^{16} x^{\frac{1}{4}}\,dx = \frac{4}{5}\left[x^{\frac{5}{4}}\right]_1^{16} = \frac{124}{5}$

$S_y = 2\pi\int_1^{16} x\sqrt{1+(y')^2}\,dx = 2\pi\int_1^{16} x^{\frac{5}{4}}\,dx = 2\pi\times\frac{4}{9}\left[x^{\frac{9}{4}}\right]_1^{16} = \frac{4088}{9}\pi$

50. $S_y = 2\pi\int_0^{\sqrt{2}} x\sqrt{1+(y')^2}\,dx = 2\pi\int_0^{\sqrt{2}} x\sqrt{1+x^2}\,dx$

$= \frac{2\pi}{3}\left[(1+x^2)^{\frac{3}{2}}\right]_0^{\sqrt{2}} = \frac{2\pi}{3}(3\sqrt{3}-1)$

51. $S_x = 2\pi\int_a^b y\sqrt{(x')^2+(y')^2}\,dt$

$= 2\pi\int_0^{\frac{\pi}{2}} \cos^2 t\sqrt{4\sin^2 t\cos^2 t+4\cos^2 t\sin^2 t}\,dt$

$= 4\sqrt{2}\pi\int_0^{\frac{\pi}{2}} \sin t\cos^3 t\,dt = -\sqrt{2}\pi\left[\cos^4 t\right]_0^{\frac{\pi}{2}} = \sqrt{2}\pi$

[다른 풀이]

주어진 매개방정식은 $x+y=\sin^2 t+\cos^2 t=1$을 만족하므로

$0\le t\le\frac{\pi}{2}$일 때, $x+y=1\,(0\le x\le 1)$로 나타낼 수 있다.

이때, $x+y=1\ (0\le x\le 1)$에 대한 선분의 중심좌표는 $\left(\frac{1}{2},\frac{1}{2}\right)$이므로

중심과 축과의 거리 $k=\frac{1}{2}$이다. 또한 선분의 길이는 $\sqrt{2}$이다.

즉, 파푸스 정리에 의하여

$S=2\pi k\times\text{길이}=2\pi\times\frac{1}{2}\times\sqrt{2}=\sqrt{2}\pi$이다.

52. $\int_0^{\pi}|v(t)|\,dt = \int_0^{\pi}|2\sin(3t)|\,dt$

$= \frac{2}{3}\int_0^{3\pi}|\sin x|\,dx = 3\times\frac{2}{3}\int_0^{\pi}\sin x\,dx = 4$

53. $\int_0^{\frac{\pi}{2}}|v(t)|\,dt = \int_0^{\frac{\pi}{2}}\sqrt{(x')^2+(y')^2}\,dt$

$= \int_0^{\frac{\pi}{2}}\sqrt{4\sin^2 t\cos^2 t+4\cos^2 t\sin^2 t}\,dt$

$= 2\sqrt{2}\int_0^{\frac{\pi}{2}}\sin t\cos t\,dt = \sqrt{2}\left[\sin^2 t\right]_0^{\frac{\pi}{2}} = \sqrt{2}$

[다른 풀이]

주어진 매개방정식은 $x+y=\sin^2 t+\cos^2 t=1$을 만족하므로

$0\le t\le\frac{\pi}{2}$일 때, $x+y=1\,(0\le x\le 1)$로 나타낼 수 있다.

따라서 직선 $x+y=1\,(0\le x\le 1)$이 움직이는 거리는 선분의 길이인 $\sqrt{2}$이다.

1. 3번	2. 4번	3. 1번	4. 2번	5. 3번
6. 1번	7. 1번	8. 4번	9. 4번	10. 2번
11. 1번	12. 1번	13. 2번	14. 3번	15. 1번
16. 3번	17. 1번	18. 2번	19. 2번	20. 3번
21. 4번	22. 1번	23. 2번	24. 3번	25. 2번

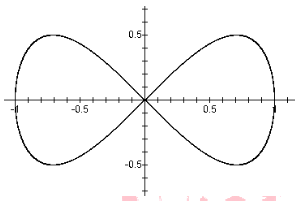
1. 두 그래프의 교점은 0과 $\frac{4}{3}$이므로 넓이는

$$S=\frac{|a-a'|}{6}(\beta-\alpha)^3=\frac{\left|\frac{1}{2}-(-1)\right|}{6}\left(\frac{4}{3}-0\right)^3=\frac{16}{27}$$ 이다.

[다른 풀이]

$$\int_0^{\frac{4}{3}}(-x^2+2x)-\frac{1}{2}x^2dx=\left[-\frac{1}{3}x^3+x^2-\frac{1}{6}x^3\right]_0^{\frac{4}{3}}$$
$$=-\frac{1}{3}\times\left(\frac{4}{3}\right)^3+\left(\frac{4}{3}\right)^2-\frac{1}{6}\times\left(\frac{4}{3}\right)^3=\frac{16}{27}$$

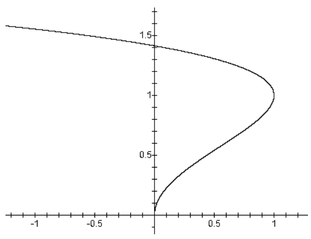
2.



따라서 $y^2=x^2-x^4 \Leftrightarrow y=\pm\sqrt{x^2-x^4}$의 그래프 중 1사분면의 넓이에 4배를 하면 둘러싸인 영역의 면적이 된다.

$$S=4\int_0^1\sqrt{x^2-x^4}\,dx=4\int_0^1 x\sqrt{1-x^2}\,dx$$
$$=4\left[-\frac{1}{3}(1-x^2)^{\frac{3}{2}}\right]_0^1=\frac{4}{3}$$

3.



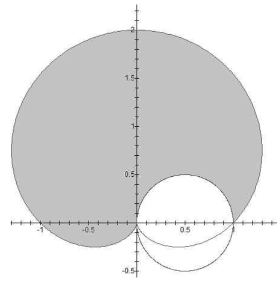
$$S=\int_0^{\sqrt{2}}x\,dy=\int_0^2(2t-t^2)\frac{1}{2\sqrt{t}}dt$$
$$=\int_0^2\sqrt{t}-\frac{1}{2}t^{\frac{3}{2}}dt=\left[\frac{2}{3}t^{\frac{3}{2}}-\frac{1}{5}t^{\frac{5}{2}}\right]_0^2$$
$$=\frac{2}{3}\times2^{\frac{3}{2}}-\frac{1}{5}\times2^{\frac{5}{2}}=\frac{8}{15}\sqrt{2}$$

[다른 풀이]

$x=2t-t^2,\ y=\sqrt{t}\ \ (t\geq0) \Leftrightarrow x=2y^2-y^4\ \ (y\geq0)$이므로 그림보면 위와 같다.

$$S=\int_0^{\sqrt{2}}x\,dy=\int_0^{\sqrt{2}}2y^2-y^4\,dy=\left[\frac{2}{3}y^3-\frac{1}{5}y^5\right]_0^{\sqrt{2}}$$
$$=\frac{2}{3}\times2^{\frac{3}{2}}-\frac{1}{5}\times2^{\frac{5}{2}}=\frac{8}{15}\sqrt{2}$$

4.



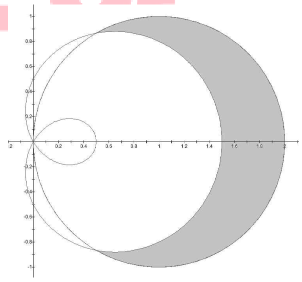
$$S=\frac{1}{2}\int_0^{\frac{3\pi}{2}}(1+\sin\theta)^2d\theta-\pi\times\left(\frac{1}{2}\right)^2\times\frac{1}{2}$$
$$=\frac{1}{2}\int_0^{\frac{3\pi}{2}}1+2\sin\theta+\sin^2\theta d\theta-\frac{\pi}{8}$$
$$=\frac{1}{2}\left(\frac{3\pi}{2}+2+3\times\frac{1}{2}\times\frac{\pi}{2}\right)-\frac{\pi}{8}$$
$$=\pi+1$$

5. $r=a\cos\theta+b$가 $(r,\theta)=\left(\frac{3}{2},\,0\right)$을 지나므로 대입하면 $\frac{3}{2}=a+b$이다.

$r=2\cos\theta$와 $r=1$의 교점은 $(r,\theta)=\left(1,\,\frac{\pi}{3}\right)$이므로 $r=a\cos\theta+b$가 지나므로 대입하면 $1=\frac{1}{2}a+b$이다.

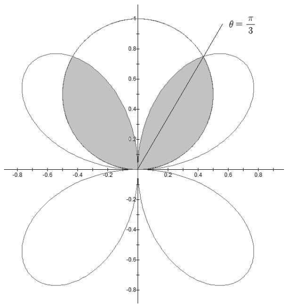
따라서 $a=1,\ b=\frac{1}{2}$이므로 $r=\cos\theta+\frac{1}{2}$이고,

$r=\frac{1}{2}+\cos\theta$의 외부와 $r=2\cos\theta$의 내부를 그려보면 아래와 같다.



$$S=2\times\frac{1}{2}\int_0^{\frac{\pi}{3}}\left[(2\cos\theta)^2-\left(\cos\theta+\frac{1}{2}\right)^2\right]d\theta$$
$$=\int_0^{\frac{\pi}{3}}\left(3\cos^2\theta-\cos\theta-\frac{1}{4}\right)d\theta$$
$$=\int_0^{\frac{\pi}{3}}\left\{\frac{3}{2}(1+\cos2\theta)-\cos\theta-\frac{1}{4}\right\}d\theta$$
$$=\left[\frac{5}{4}\theta+\frac{3}{4}\sin2\theta-\sin\theta\right]_0^{\frac{\pi}{3}}=\frac{5}{12}\pi-\frac{\sqrt{3}}{8}$$

6.



$$S=2\times\left[\frac{1}{2}\int_{0}^{\frac{\pi}{3}}\sin^2\theta\,d\theta+\frac{1}{2}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\sin^2 2\theta\,d\theta\right]$$

$$=\int_{0}^{\frac{\pi}{3}}\frac{1-\cos2\theta}{2}\,d\theta+\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\frac{1-\cos4\theta}{2}\,d\theta$$

$$=\frac{1}{2}\left[\theta-\frac{1}{2}\sin2\theta\right]_{0}^{\frac{\pi}{3}}+\frac{1}{2}\left[\theta-\frac{1}{4}\sin4\theta\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

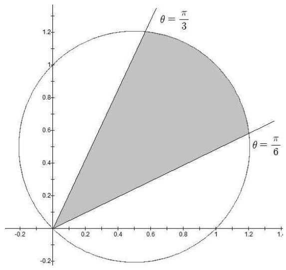
$$=\frac{1}{2}\left(\frac{\pi}{3}-\frac{1}{2}\times\frac{\sqrt{3}}{2}\right)+\frac{1}{2}\left(\frac{\pi}{2}-\frac{\pi}{3}-\frac{1}{4}\times\frac{\sqrt{3}}{2}\right)$$

$$=\frac{\pi}{4}-\frac{3\sqrt{3}}{16}$$

7. $x^2+y^2=x+y$

$\Rightarrow\ r^2=r\cos\theta+r\sin\theta\ \ \Rightarrow\ r=\cos\theta+\sin\theta$

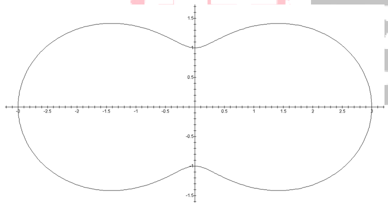
이므로 그림을 그려보면 아래와 같다.



$$S=\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}(\cos\theta+\sin\theta)^2d\theta=\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}1+2\sin\theta\cos\theta\,d\theta$$

$$=\frac{1}{2}\left[\theta+\sin^2\theta\right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}=\frac{\pi}{12}+\frac{1}{4}$$

8. $r=2+\cos^2\theta-\sin^2\theta=2+\cos2\theta$ 의 그래프는 다음과 같다.



$$S=4\times\frac{1}{2}\int_{0}^{\frac{\pi}{2}}(2+\cos2\theta)^2d\theta$$

$$=2\int_{0}^{\frac{\pi}{2}}(4+4\cos2\theta+\cos^2 2\theta)d\theta$$

$$=2\left[4\theta+2\sin2\theta\right]_{0}^{\frac{\pi}{2}}+\int_{0}^{\pi}\cos^2t\,dt\ \ (\because 2\theta=t)$$

$$=4\pi+2\times\frac{1}{2}\times\frac{\pi}{2}=\frac{9}{2}\pi$$

9. $r=2+\sin\theta+\cos\theta=2+\sqrt{2}\sin\left(\theta+\frac{\pi}{4}\right)$의 그래프는

$r=2+\sqrt{2}\sin\theta$의 그래프를 시계방향으로 $\frac{\pi}{4}$만큼 회전한 모양이다.

따라서 넓이는 $r=2+\sqrt{2}\sin\theta$와 동일하며

$S=\frac{\pi}{2}(2a^2+b^2)=\frac{\pi}{2}(2\times4+2)=5\pi$이다.

[다른 풀이]

$$S=2\times\frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(2+\sqrt{2}\sin\theta)^2d\theta$$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}4+4\sqrt{2}\sin\theta+2\sin^2\theta\,d\theta$$

$$=4\pi+0+2\times2\times\frac{1}{2}\times\frac{\pi}{2}$$

$$=5\pi$$

10. $$l=\int_{1}^{e^4}\sqrt{1+(y')^2}\,dx=\int_{1}^{e^4}\sqrt{1+\left(2x-\frac{1}{8x}\right)^2}\,dx$$

$$=\int_{1}^{e^4}\sqrt{4x^2+\frac{1}{2}+\frac{1}{64x^2}}\,dx=\int_{1}^{e^4}\sqrt{\left(2x+\frac{1}{8x}\right)^2}\,dx$$

$$=\int_{1}^{e^4}2x+\frac{1}{8x}dx=\left[x^2+\frac{1}{8}\ln x\right]_{1}^{e^4}=e^8-\frac{1}{2}$$

11. $$l=\int_{0}^{1}\sqrt{(x')^2+(y')^2+(z')^2}\,dt$$

$$=\int_{0}^{1}\sqrt{(2t)^2+(-\sin t)^2+(\cos t)^2}\,dt$$

$$=\int_{0}^{1}\sqrt{4t^2+1}\,dt$$

$$=\frac{1}{2}\int_{0}^{\tan^{-1}2}\sec^3\theta\,d\theta\ \ (\because 2t=\tan\theta)$$

$$=\frac{1}{2}\times\frac{1}{2}\left[\sec\theta\tan\theta+\ln(\sec\theta+\tan\theta)\right]_{0}^{\tan^{-1}2}$$

$$=\frac{1}{4}\left[2\sqrt{5}+\ln(\sqrt{5}+2)\right]$$

12. $0\le\theta\le\frac{\pi}{2}$ 일 때, 극곡선 $r=1-\sin\theta$ 의 길이는

$(4-2\sqrt{2})a=(4-2\sqrt{2})\times1$ 이다.

[다른 풀이1]

$$l=\int_{0}^{\frac{\pi}{2}}\sqrt{r^2+(r')^2}\,d\theta=\int_{0}^{\frac{\pi}{2}}\sqrt{(1-\sin\theta)^2+\cos^2\theta}\,d\theta$$

$$=\int_{0}^{\frac{\pi}{2}}\sqrt{2-2\sin\theta}\,d\theta$$

$$=\int_{0}^{\frac{\pi}{2}}\sqrt{2-2\cos t}\,dt\ \left(\because \theta=\frac{\pi}{2}-t\right)$$

$$=\int_{0}^{\frac{\pi}{2}}\sqrt{2^2\sin^2\frac{t}{2}}\,dt$$

$$=\int_{0}^{\frac{\pi}{2}}2\sin\frac{t}{2}dt=\left[-4\cos\frac{t}{2}\right]_{0}^{\frac{\pi}{2}}=4-2\sqrt{2}$$

[다른 풀이2]

$0\le\theta\le\frac{\pi}{2}$ 일 때, 극곡선 $r=1-\sin\theta$ 의 길이는

$0\le\theta\le\frac{\pi}{2}$ 일 때, 극곡선 $r=1-\cos\theta$ 의 길이와 동일하다.

$$l=\int_{0}^{\frac{\pi}{2}}\sqrt{r^2+(r')^2}\,d\theta=\int_{0}^{\frac{\pi}{2}}\sqrt{(1-\cos\theta)^2+\sin^2\theta}\,d\theta$$

$$=\int_{0}^{\frac{\pi}{2}}\sqrt{2-2\cos\theta}\,d\theta=\int_{0}^{\frac{\pi}{2}}\sqrt{2^2\sin^2\frac{\theta}{2}}\,d\theta$$

$$=\int_{0}^{\frac{\pi}{2}}2\sin\frac{\theta}{2}d\theta=\left[-4\cos\frac{\theta}{2}\right]_{0}^{\frac{\pi}{2}}=4-2\sqrt{2}$$

13. $x^2+y^2=\sqrt{x^2+y^2}+x$

$\Rightarrow\ r^2=r+r\cos\theta$

$\Rightarrow\ r=1+\cos\theta$

이므로 곡선의 길이는 $2\sqrt{2}a=2\sqrt{2}\times1$ 이다.

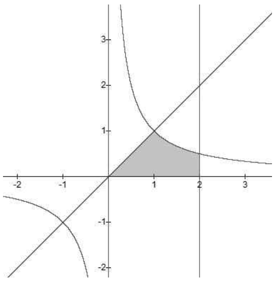
[다른 풀이]

$$l=\int_{0}^{\frac{\pi}{2}}\sqrt{r^2+(r')^2}\,d\theta=\int_{0}^{\frac{\pi}{2}}\sqrt{(1+\cos\theta)^2+\sin^2\theta}\,d\theta$$

$$=\int_{0}^{\frac{\pi}{2}}\sqrt{2+2\cos\theta}\,d\theta=\int_{0}^{\frac{\pi}{2}}\sqrt{2^2\cos^2\frac{\theta}{2}}\,d\theta$$

$$=\int_{0}^{\frac{\pi}{2}}2\cos\frac{\theta}{2}d\theta=\left[4\sin\frac{\theta}{2}\right]_{0}^{\frac{\pi}{2}}=2\sqrt{2}$$

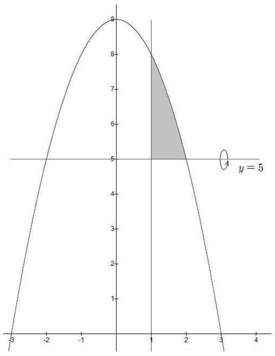
14.



$V_x = \pi\int_0^1 x^2 dx + \pi\int_1^2 \frac{1}{x^2} dx$

$= \pi \times \frac{1}{3} + \pi\left[-\frac{1}{x}\right]_1^2 = \frac{5\pi}{6}$

15.

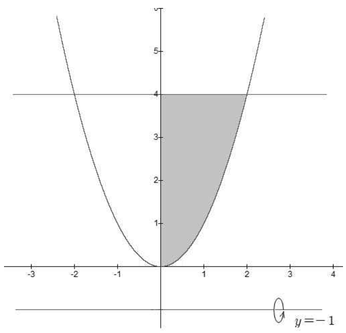


$V = \pi\int_1^2 (y-5)^2 dx = \pi\int_1^2 (9-x^2-5)^2 dx$

$= \pi\int_1^2 (4-x^2)^2 dx = \pi\int_1^2 16-8x^2+x^4 dx$

$= \pi\left[16x - \frac{8}{3}x^3 + \frac{1}{5}x^5\right]_1^2 = \frac{53}{15}\pi$

16.

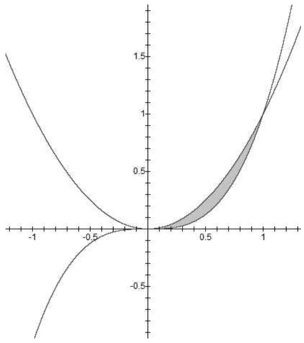


$V = \pi\int_0^2 5^2 - (y+1)^2 dx = \pi\int_0^2 25 - (x^2+1)^2 dx$

$= \pi\int_0^2 24 - 2x^2 - x^4 dx = \pi\left[24x - \frac{2}{3}x^3 - \frac{1}{5}x^5\right]_0^2$

$= \pi\left(48 - \frac{16}{3} - \frac{32}{5}\right) = \frac{544}{15}\pi$

17.



y축 중심으로 회전한 부피는 다음과 같다.

$V = 2\pi\int_0^1 x(y_1 - y_2)dx = 2\pi\int_0^1 x(x^2 - x^3)dx$

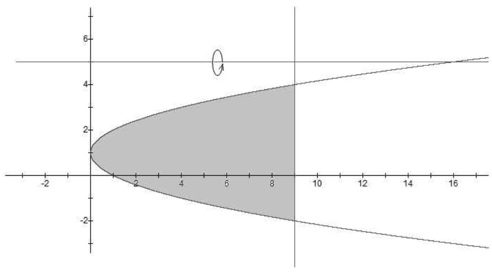
$= 2\pi\int_0^1 x^3 - x^4 dx = 2\pi\left[\frac{1}{4}x^4 - \frac{1}{5}x^5\right]_0^1 = 2\pi\left(\frac{1}{4} - \frac{1}{5}\right) = \frac{\pi}{10}$

x축 중심으로 회전한 부피는 다음과 같다.

$V_x = \pi\int_0^1 y_1^2 - y_2^2 dx = \pi\int_0^1 x^4 - x^6 dx = \pi\left[\frac{1}{5}x^5 - \frac{1}{7}x^7\right]_0^1$

$= \pi\left(\frac{1}{5} - \frac{1}{7}\right) = \frac{2\pi}{35}$

18.



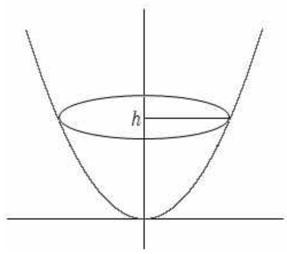
영역의 중심좌표는 $M(*, 1)$이므로 축과의 거리는 $k = 4$이다.

또한 영역의 넓이는 $\frac{|a|}{6}(\beta - \alpha)^3 = \frac{1}{6}(4 - (-2))^3 = 36$이다.

파푸스 정리에 의하여 회전체의 부피는

$V = 2\pi k \times \text{넓이} = 2\pi \times 4 \times 36 = 288\pi$ 이다.

19.



$\frac{dV}{dt} = 2,\ h = 9$일 때,

$V = \pi\int_0^h x^2 dy = \pi\int_0^h y dy$이므로 $\frac{dV}{dt} = \pi h \times \frac{dh}{dt}$이다.

즉, $2 = 9\pi\frac{dh}{dt}$이므로 $\frac{dh}{dt} = \frac{2}{9\pi}(cm/\sec)$이다.

20. $V = \int_{-1}^1 \text{단면의 넓이}\, dy = \int_{-1}^1 \pi x^2 dy$

$= \pi\int_{-1}^1 \frac{1}{4}(1-y^2)^2 dy = \frac{\pi}{4}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^5\theta\, d\theta \quad (\because y = \sin\theta)$

$= \frac{\pi}{4} \times 2 \times \frac{4}{5} \times \frac{2}{3} = \frac{4}{15}\pi$

21. $S_x = 2\pi\int_1^2 y\sqrt{1+(x')^2}\,dy = 2\pi\int_1^2 y\sqrt{1+\left(\frac{1}{2}y^3-\frac{1}{2}y^{-3}\right)^2}\,dy$

$= 2\pi\int_1^2 y\sqrt{\frac{1}{4}y^6+\frac{1}{2}+\frac{1}{4}y^{-6}}\,dy = 2\pi\int_1^2 y\sqrt{\left(\frac{1}{2}y^3+\frac{1}{2}y^{-3}\right)^2}\,dy$

$= \pi\int_1^2 y^4+y^{-2}dy = \pi\left[\frac{1}{5}y^5-\frac{1}{y}\right]_1^2 = \frac{67}{10}\pi$

22. $x+y=1$ $(0 \le x \le 1)$에 대한 선분의 중심좌표는 $\left(\frac{1}{2}, \frac{1}{2}\right)$이므로

중심과 축과의 거리 $k=\frac{1}{2}$이다. 또한 선분의 길이는 $\sqrt{2}$이다.

즉, 파푸스 정리에 의하여

$S = 2\pi k \times$ 길이 $= 2\pi\times\frac{1}{2}\times\sqrt{2} = \sqrt{2}\pi$이다.

[다른 풀이]

$S_x = 2\pi\int_0^{\frac{\pi}{2}} y\sqrt{(x')^2+(y')^2}\,dt$

$= 2\pi\int_0^{\frac{\pi}{2}} \sin^2 t\sqrt{(-2\cos t\sin t)^2+(2\sin t\cos t)^2}\,dt$

$= 4\sqrt{2}\pi\int_0^{\frac{\pi}{2}} \sin^3 t\cos t dt$

$= 4\sqrt{2}\pi\times\frac{1}{4}\left[\sin^4 t\right]_0^{\frac{\pi}{2}} = \sqrt{2}\pi$

23. $S_x = 2\pi\int_0^{\frac{\pi}{2}} y\sqrt{r^2+(r')^2}\,d\theta = 2\pi\int_0^{\frac{\pi}{2}} r\sin\theta\sqrt{r^2+(r')^2}\,d\theta$

$= 2\pi\int_0^{\frac{\pi}{2}} e^{\theta}\sin\theta\sqrt{e^{2\theta}+e^{2\theta}}\,d\theta$

$= 2\sqrt{2}\pi\int_0^{\frac{\pi}{2}} e^{2\theta}\sin\theta d\theta$

$= 2\sqrt{2}\pi\left[\frac{e^{2\theta}}{5}(2\sin\theta-\cos\theta)\right]_0^{\frac{\pi}{2}}$

$= \frac{2\sqrt{2}}{5}\pi(2e^{\pi}+1)$

24. $S = 4\pi r^2 \times \frac{\text{구간의 길이}}{\text{지름}}$

$= 4\pi\times 9\times\frac{2-(-1)}{6} = 18\pi$

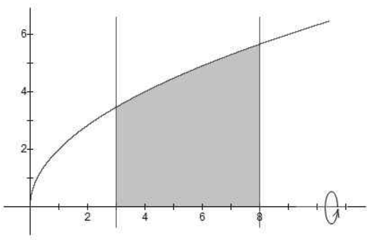
[다른 풀이]

$S_x = 2\pi\int_{-1}^2 y\sqrt{1+(y')^2}\,dx$

$= 2\pi\int_{-1}^2 \sqrt{9-x^2}\sqrt{1+\left(\frac{-x}{\sqrt{9-x^2}}\right)^2}\,dx$

$= 2\pi\int_{-1}^2 3dx = 2\pi\times 3\times 3 = 18\pi$

25.



곡선을 회전하여 나온 겉넓이는 다음과 같다.

$S_1 = 2\pi\int_3^8 y\sqrt{1+(y')^2}\,dx$

$= 2\pi\int_3^8 2\sqrt{x}\sqrt{1+\frac{1}{x}}\,dx$

$= 4\pi\int_3^8 \sqrt{x+1}\,dx = 4\pi\left[\frac{2}{3}(x+1)^{\frac{3}{2}}\right]_3^8$

$= \frac{152}{3}\pi$

양 끝의 표면은 원의 넓이와 같다.

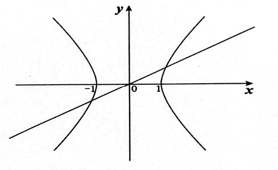
$S_2 = \pi\times(2\sqrt{3})^2 = 12\pi$, $S_3 = \pi\times(2\sqrt{8})^2 = 32\pi$

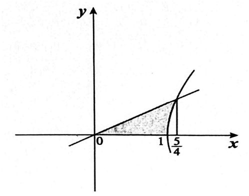
즉, 영역을 회전하여 나온 회전체의 겉넓이는 $S_1+S_2+S_3 = \frac{284}{3}\pi$

이다.

1. 1번	2. 2번	3. 5번	4. 1번	5. 2번
6. 1번	7. 4번	8. 2번	9. 2번	10. 3번
11. 3번	12. 4번	13. 2번	14. 4번	15. 1번
16. 3번	17. 4번			

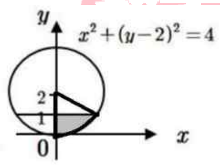
1. $\begin{cases} x^2-y^2=1 \\ 5y=3x \end{cases}$의 두 그래프를 그려보면,





$$A=\frac{1}{2}\times\frac{3}{4}\times\frac{5}{4}-\int_1^{\frac{5}{4}} ydx$$
$$=\frac{15}{32}-\int_0^{\ln 2}\sinh^2 t dt \quad (\because x=\cosh t,\ y=\sinh t)$$
$$=\frac{15}{32}-\int_0^{\ln 2}\frac{1}{4}(e^{2t}-2+e^{-2t})dt$$
$$=\frac{15}{32}-\frac{1}{4}\left[\frac{1}{2}e^{2t}-2t-\frac{1}{2}e^{-2t}\right]_0^{\ln 2}$$
$$=\frac{15}{32}-\frac{1}{4}\left(\frac{1}{2}e^{2\ln 2}-2\ln 2-\frac{1}{2}e^{-2\ln 2}\right)=\frac{1}{2}\ln 2$$

2.



주어진 영역의 면적은 중심각이 $\frac{\pi}{3}$인 부채꼴의 면적에서 작은 삼각형의 면적을 빼면 된다.

$$\therefore\ \frac{1}{2}\times 2^2\times\frac{\pi}{3}-\sqrt{3}\times 1\times\frac{1}{2}=\frac{2\pi}{3}-\frac{\sqrt{3}}{2}$$

3. $2\int_{\frac{\pi}{4}}^{0}|\cos 2t\tan t|(-2\sin 2t)dt=4\int_0^{\frac{\pi}{4}}\sin 2t\cos 2t\tan t dt$

$$=4\int_0^{\frac{\pi}{4}}(2\sin t\cos t)(1-2\sin^2 t)\times\frac{\sin t}{\cos t}dt$$
$$=8\int_0^{\frac{\pi}{4}}\sin^2 t(1-2\sin^2 t)dt=8\int_0^{\frac{\pi}{4}}(\sin^2 t-2\sin^4 t)dt$$

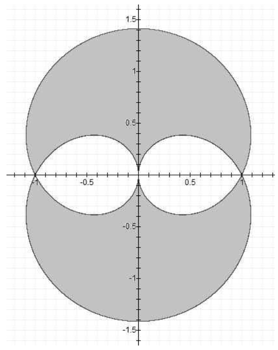
1) $\int_0^{\frac{\pi}{4}}\sin^2 t dt=\left[\frac{t}{2}-\frac{1}{4}\sin 2t\right]_0^{\frac{\pi}{4}}=\frac{\pi}{8}-\frac{1}{4}$

2) $\int_0^{\frac{\pi}{4}}\sin^4 dt=\int_0^{\frac{\pi}{4}}\left(\frac{1-\cos 2t}{2}\right)^2 dt=\int_0^{\frac{\pi}{4}}\frac{1}{4}(1-2\cos 2t+\cos^2 2t)dt$

$$=\frac{1}{4}\left[t-\sin 2t+\frac{t}{2}+\frac{1}{8}\sin 4t\right]$$
$$=\frac{1}{4}\left(\frac{\pi}{4}-1+\frac{\pi}{8}\right)=\frac{1}{4}\left(\frac{3}{8}\pi-1\right)$$

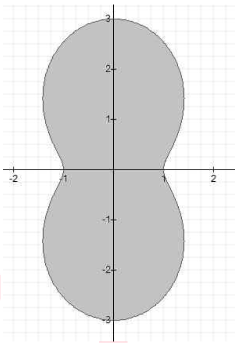
즉, $A=8\left\{\frac{\pi}{8}-\frac{1}{4}-2\frac{1}{4}\left(\frac{3}{8}\pi-1\right)\right\}=2-\frac{\pi}{2}$

4.



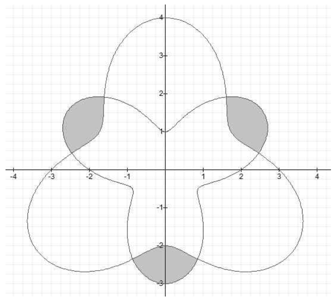
$$4\times\left(\frac{1}{2}\int_0^{\frac{\pi}{2}}1+\sin\theta d\theta-\frac{1}{2}\int_{\pi}^{\frac{3\pi}{2}}1+\sin\theta d\theta\right)$$
$$=2\left([\theta-\cos\theta]_0^{\frac{\pi}{2}}-[\theta-\cos\theta]_{\pi}^{\frac{3\pi}{2}}\right)=4$$

5.



$$\frac{1}{2}\int_0^{2\pi}(2-\cos 2\theta)^2 d\theta$$
$$=\frac{1}{2}\int_0^{2\pi}4-4\cos 2\theta+\cos^2 2\theta\, d\theta$$
$$=\frac{1}{2}\int_0^{2\pi}\frac{9}{2}-4\cos 2\theta+\frac{1}{2}\cos 4\theta\, d\theta$$
$$=\frac{1}{2}\left[\frac{9}{2}\theta-2\sin 2\theta+\frac{1}{8}\sin 4\theta\right]_0^{2\pi}=\frac{9\pi}{2}$$

6.



$$\left[\frac{1}{2}\int_{\frac{\pi}{18}}^{\frac{5}{18}\pi}\{(2+\sin 3\theta)^2-(3-\sin 3\theta)^2\}d\theta\right]\times 3$$
$$=\frac{3}{2}\int_{\frac{\pi}{18}}^{\frac{5}{18}\pi}\{4+4\sin 3\theta+\sin^2 3\theta-(9-6\sin 3\theta+\sin^2 3\theta)\}d\theta$$
$$=\frac{3}{2}\int_{\frac{\pi}{18}}^{\frac{5}{18}\pi}(-5+10\sin 3\theta)d\theta=\frac{3}{2}\left[-5\theta-\frac{10}{3}\cos 3\theta\right]_{\frac{\pi}{18}}^{\frac{5}{18}\pi}$$
$$=5\sqrt{3}-\frac{5}{3}\pi$$

7. $y=\sqrt[3]{2-x^3} \;\Leftrightarrow\; r=\dfrac{\sqrt[3]{2}}{(\cos^3\theta+\sin^3\theta)^{\frac{1}{3}}}$ 이므로

$s(t)=\dfrac{1}{2}\displaystyle\int_0^{\tan^{-1}t}\dfrac{2^{\frac{2}{3}}}{(\cos^3\theta+\sin^3\theta)^{\frac{2}{3}}}\,d\theta$ 이다.

$s'(t)=\dfrac{1}{\sqrt[3]{2}}\dfrac{1}{\{\cos^3(\tan^{-1}t)+\sin^3(\tan^{-1}t)\}^{\frac{2}{3}}}\times\dfrac{1}{1+t^2}$

$\therefore\; s'(1)=\dfrac{1}{2}$

8. $r=\dfrac{2}{2+\cos\theta} \;\Leftrightarrow\; 2\sqrt{x^2+y^2}=2-x$

$\Leftrightarrow\; 3x^2+4x+4y^2=4 \;\Leftrightarrow\; \dfrac{\left(x+\frac{2}{3}\right)^2}{\left(\frac{4}{3}\right)^2}+\dfrac{y^2}{\left(\frac{2}{\sqrt{3}}\right)^2}=1$

타원의 방정식의 면적 : $\dfrac{2}{\sqrt{3}}\times\dfrac{4}{3}\times\pi=\dfrac{8\pi}{3\sqrt{3}}$

9. $L=\displaystyle\int_0^{\frac{\pi}{3}}\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$

$=\displaystyle\int_0^{\frac{\pi}{3}}\sqrt{(-\sin t)^2+(-2\sin 2t)^2}\,dt$

$=\displaystyle\int_0^{\frac{\pi}{3}}\sin t\sqrt{1+16\cos^2 t}\,dt=\int_{\frac{1}{2}}^{1}\sqrt{1+16x^2}\,dx$

이므로 하합 : $\dfrac{\sqrt{5}}{2}$, 상합 : $\dfrac{\sqrt{17}}{2}$ 이다.

$\therefore\; \dfrac{\sqrt{5}}{2}\le L\le\dfrac{\sqrt{17}}{2}$

10. $x^2+(y-a)^2\le 1-\dfrac{1}{4}a \;\Leftrightarrow\;$ 중심 : $(0,a)$, 반지름 : $\sqrt{1-\dfrac{1}{4}a}$

Pappus 정리를 이용하면 $V_a=\left(1-\dfrac{1}{4}a\right)\times 2\pi\times a=2\pi\left(a-\dfrac{1}{4}a^2\right)$이다.

$V'_a=2\pi\left(1-\dfrac{1}{2}a\right)$이므로 $a=2$일 때, 부피는 최댓값을 갖는다.

11. Pappus 정리를 이용하면 $V=2\pi\times\pi r\times 3\pi r^2=6\pi^3r^3$이다.

12. $V_1=\displaystyle\int_0^1\pi\{1^2-(\sqrt[4]{x})^2\}dx=\pi\int_0^1 1-\sqrt{x}\,dx=\frac{\pi}{3}$

$V_2=\pi\displaystyle\int_0^1\{y^2-(y^4)^2\}dy=\frac{2}{9}\pi$

이므로 $V_1 : V_2=\dfrac{\pi}{3}:\dfrac{2\pi}{9}=3:2$ 이다.

13. $V_h=\pi\displaystyle\int_0^1 r^2dh=\pi\int_0^1(1-h)^4dh=\frac{\pi}{5}$

$\therefore\; \dfrac{\pi}{3}-\dfrac{\pi}{5}=\dfrac{2\pi}{15}$

14. $1\le x\le 3$에 대하여 $x^2+y^2=16$을 x축 중심으로 회전한 부피와 같다.

$V_x=\pi\displaystyle\int_1^3 y^2dx=\pi\int_1^3 16-x^2dx$

$=\pi\left[16x-\dfrac{1}{3}x^3\right]_1^3=\dfrac{70}{3}\pi$

15. $V=\pi\displaystyle\int_c^d x^2dy=\pi\int_0^4 t^3\times\frac{1}{2}t^{-\frac{1}{2}}dt$

$=\dfrac{\pi}{2}\displaystyle\int_0^4 t^{\frac{5}{2}}dt=\frac{\pi}{2}\times\frac{2}{7}\left[t^{\frac{7}{2}}\right]_0^4=\frac{2^7}{7}\pi$

16. $r=2\sin\theta,\ -\dfrac{\pi}{4}\le\theta\le\dfrac{\pi}{4}$에 대하여 x축 회전한 겉넓이와 같다.

$S_x=2\pi\displaystyle\int_c^d r\sin\theta\sqrt{r^2+(r')^2}\,d\theta$

$=2\pi\displaystyle\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}2\sin^2\theta\sqrt{4\sin^2\theta+4\cos^2\theta}\,d\theta$

$=16\pi\displaystyle\int_0^{\frac{\pi}{4}}\sin^2\theta d\theta=8\pi\int_0^{\frac{\pi}{4}}1-\cos 2\theta d\theta$

$=8\pi\left[\theta-\dfrac{1}{2}\sin 2\theta\right]_0^{\frac{\pi}{4}}=2\pi(\pi-2)$

17. $S_x=2\pi\displaystyle\int_c^d r\sin\theta\sqrt{r^2+(r')^2}\,d\theta$

$=2\pi\displaystyle\int_0^{\frac{\pi}{2}}(1+\cos\theta)\sin\theta\sqrt{(1+\cos\theta)^2+\sin^2\theta}\,d\theta$

$=2\sqrt{2}\pi\displaystyle\int_0^{\frac{\pi}{2}}\sin\theta(1+\cos\theta)^{\frac{3}{2}}d\theta$

$=-2\sqrt{2}\pi\times\dfrac{2}{5}\left[(1+\cos\theta)^{\frac{5}{2}}\right]_0^{\frac{\pi}{2}}=\dfrac{4\sqrt{2}}{5}\pi(4\sqrt{2}-1)$